

A Test of Learning Progress Models Using an AI-Enabled Knowledge Representation System

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Abstract: This study tested two competing learning progress models - the three-stage and two-stage models - based on an AI-enabled formative assessment tool. Additionally, human-rated scores were used to further validate the models. The data for this investigation consisted of expository essays about a complex problem scenario written by 116 students and 6 experts. The validation analyses, including descriptive statistics, C-LCDM, and group-mean difference tests, demonstrated that the two-stage model was a better framework given the technology. The findings showed the potential of the model to determine learners' conceptual change over time.

Introduction

From an expertise development standpoint, learning is a developmental process toward expert-like problem-solving knowledge and skills (Alexander, 2004; Chi, 2006). Kim (2012, 2015) built on expertise development theories to conceptualize and test learning progress models, shifting the focus of development from long-term change to short-term change in task-level problem-solving in the classroom context. Kim (2012) theorized that learner mental models could represent different levels of understanding of a problem situation, and thus tracking mental model changes could explain how learners develop expertise. A subsequent study proposed computational techniques to extract a knowledge structure from a student's expository essay and leveraged a specific technique, the Continuous Log-Linear Cognitive Diagnostic Model (C-LCDM) method (Bozard, 2010), to test levels of mental models (Kim, 2015).

In this study, we use an AI-based assessment system called SMART (Student Mental Model Analyzer for Research and Teaching) to extract mental models from students' essays and test two competing learning progress models. Since different modeling techniques can produce varying concept maps, it is important to validate relevant diagnostic models. SMART's diagnostic model can monitor students' learning growth over time. The study aims to answer the following questions:

- How do the models explain the stages of learning progress?
- How closely do the computerized diagnostic results align with human judgments?

Learning progress models

Learners' problem-solving abilities can be reflected in their knowledge structure, which is necessary for solving complex problems. Assessing students' problem-solving skills requires examining their knowledge base, which includes ideas and concepts learned from the problem situation (Gijbel et al., 2005). Scholars have introduced three dimensions to understand knowledge structure, known as the 3S dimensions (Kim, 2012): (a) the surface dimension, which concerns the number of constituents of a mental model (e.g., concepts and their relations), (b) the structural dimension, which describes how those components are connected and organized in the knowledge structure (e.g., size and cohesion), and (c) the semantic dimension, which reflects the extent to which students include underlying ideas and concepts used by experts. We use the 3S dimensions to characterize stages of two competing models (i.e., three-stage and two-stage models) as detailed in Table 1. The three-stage learning progress model is based on the Model of Domain Learning (Alexander, 2004) and conceptualizes three stages: acclimation (the initial stage in which most learners experience a lack of pre-structural knowledge), competent (in which learners reconsider and repair their irrelevant preconceptions through teaching and learning), and proficiency-expertise (in which proficient learners construct a sufficient knowledge structure reflecting the problem situation). The two-stage model, called conceptual change, explains that for complex problems, conceptual change from a naïve knowledge model to an informed problem space often requires a shift in fundamental knowledge structure, especially (Chi, 2008).

Methods

Research context and data source

This study collected data from 116 undergraduate students and six professors in the field of instructional sciences. The students were asked to provide their evaluation on the potential factors and variables that led to the failure of a technology innovation project: the implementation of tablet PCs in high-school classrooms that did not result in significant improvements in classroom practices and student performance. The six professors collaborated to develop a reference response using a Delphi research process. To analyze the written essays, the researchers utilized a knowledge-based AI system called SMART. The SMART system performs various core analytics tasks, such as extracting text variables from the essay, eliciting a concept map, abstracting the key knowledge structure from the concept map, comparing the student's concept map to an expert's concept map, and generating multidimensional similarity measures.

Table 1
Competing Learning Progress Models

Three Stage Model	Description	Two Stage Model	Description
Acclimation	(a) All dimensions (surface, structural, and semantic) are quite dissimilar to expert models; or (b) knowledge structures have similar surface dimension to expert models but are missing some structural and semantic dimensions.	Misconception (Novice)	(a) All dimensions (surface, structural, and semantic) are quite dissimilar to expert models; (b) knowledge structures have similar surface dimension to expert models but are missing some structural and semantic dimensions; or (c) Structural dimension might appear to be mastered because mental models, which consist of a small amount of contextual and abstract knowledge, are likely to look cohesive and connected.
Competence	(a) Structural dimension might appear to be mastered because mental models, which consist of a small amount of contextual and abstract knowledge, are likely to look cohesive and connected; or (b) student and expert models are highly similar in semantic dimension but dissimilar in surface and structural dimensions.	Conception (Expert)	(a) Structural dimension shows sufficient complexity while surface dimension is adequate, but not enough to guarantee a semantic fit; (b) knowledge structures are well-featured in all dimensions (surface, structure, and semantic); (c) a significant number of principles (semantic) create a cohesive structure (structural) but with a small number of concepts (surface); or (d) student and expert models are highly similar in semantic dimension but dissimilar in surface and structural dimensions.
Proficiency-Expertise	(a) Structural dimension shows sufficient complexity while surface dimension is adequate, but not enough to guarantee a semantic fit; (b) knowledge structures are well-featured in all dimensions (surface, structure, and semantic); or (c) a significant number of principles (semantic) create a cohesive structure (structural) but with a small number of concepts (surface).		

Data preparation

The SMART analytics system generates concept map indices and knowledge components. Model comparisons result in similarity measures (ranging from 0 to 1) that fall under one of the 3S dimensions, as outlined in Table 2. These hypothetical relationships between the similarity measures and the 3S dimensions were validated by Kim (2015). To further validate the results, two human evaluators used a rubric (Gao et al., 2019) to score the learner responses on four dimensions: content quality, content coverage, content coherence, and argument quality, each of which was rated on a six-point scale. The interrater reliability scores ranged from 0.87 to 0.91 (acceptable ICCs > 0.80; Graham et al., 2012).

Data analysis

We utilized the Continuous Log-Linear Cognitive Diagnostic Model (C-LCDM) method (Bozard, 2010) to identify the stages of mental models and compare the two competing learning progress models. The 3S dimensions are attribute variables in C-LCDM, and each dimension can be considered mastered when its probability is above 0.5 (Rupp et al., 2010). Therefore, the three dimensions ($k = 3$) create eight classes (2^k classes). The mastery profiles of attributes for individuals determine their latent class membership.

Table 2
Model-Based Similarity Measures

Similarity Measure	Definition	3S structure
Number of concepts	Compare the number of concepts (nodes) in two models	Surface
Number of relations	Compare the number of links (edges) in two models	Surface
Density of graphs	Compare the density of two models	Surface
Mean Distance	Compare the mean distances in two models	Structure
Diameter	Compare the largest geodesics in two models	Structure
Concept Matching	Compare semantically identical concepts	Semantic
Propositional Matching	Compare fully identical propositions (edges) between two concept maps	Semantic
Recall-C	The proportion of key concepts that appear in a student summary	Semantic
Recall-P	The proportion of key relations that appear in a student summary	Semantic

Findings

The results revealed the presence of four latent classes: classes 1, 3, 5, and 7, which represented 17%, 31%, 12%, and 40% of the total latent classes, respectively. According to the two learning progress models, these classes exhibited acclimation (29%), competence (31%), and proficiency (41%) in the three-stage model, and misconception (60%) and conception (40%) in the two-stage model (as shown in Table 3). However, contrary to its theoretical assumption, the stages were arranged in descending order, with the second stage of the three-stage model exhibiting lower values in most similarity measures than the first stage. In contrast, the two-stage model displayed an increase in all similarity measures from the misconception to the conception stage.

Table 3
The Estimated Final Class Counts and Proportions

Latent Class	3S Attribute			Three Stages	Two Stage	Estimated Classification (Count)
	S1	S2	S3			
Class 1 (C_1)	0	0	0	Acclimation	Misconception	0.171 (19)
Class 2 (C_2)	0	0	1	Competence	Conception	-
Class 3 (C_3)	0	1	0	Competence	Misconception	0.308 (36)
Class 4 (C_4)	0	1	1	Proficiency	Conception	-
Class 5 (C_5)	1	0	0	Acclimation	Misconception	0.120 (14)
Class 6 (C_6)	1	0	1	-	-	-
Class 7 (C_7)	1	1	0	Proficiency	Conception	0.402 (47)
Class 8 (C_8)	1	1	1	Proficiency	Conception	-

Note. $N = 116$. 3S attributes involve the three dimensions of knowledge structure: Surface (S1), Structural (S2), and Semantic (S3). 0 = absent/non-mastered and 1 = present/mastered.

We conducted an analysis to determine if the two learning progress models could differentiate human-rated scores across the four criteria outlined in Table 4. The results of all group-mean difference tests indicated significant differences between the stages in terms of rubric scores. Consistent with our previous findings, the second stage of the three-stage model received lower scores compared to the first stage. In contrast, the two-stage model aligns with the theoretical assumption, as evidenced by increasing rubric scores from the misconception to the conception stage.

Discussion

An AI-enabled knowledge representation technology can directly influence the way knowledge structures are modeled from students' textual explanations. In this study, the AI-enabled assessment setting demonstrated the potential of the two-stage learning progress model (conceptual change) to characterize the stages of mental models regarding complex problem situations. For example, some students rely on deeply entrenched naive knowledge

that is incorrect (misconception), while others leverage sufficient prior knowledge connected to a given problem situation (conception). Pedagogically, the two-stage model, along with the proposed technology-based measurement, can detect students' conceptual change when conceptualizing complex problems. Instructors can use this model to monitor individual students' learning progress and determine when and how to provide instructional remedies, or when to move on to the next assignment. This initial work opens pathways to future studies, and the suggested models and methods should be tested in various technology-based knowledge assessment environments with problems. For example, well-structured problems in closed-world domains or source-based writing tasks (summarization) may produce a new dynamic in the 3S dimensions and a learning progress model.

Table 4
Means, Standard Deviations, Spearman's rho, and Group Difference Statistics for Rubric Criteria

	Three-Stage Model			F	Sig.	Two-Stage Model		F	Sig.
	Stage1	Stage2	Stage3			Stage1	Stage2		
Content	2.338	1.861	3.152	21.79	.00	2.093	3.152	37.35	.00
Quality	(1.133)	(.743)	(.809)			(.975)	(.809)		
Content	2.132	1.931	2.663	7.61	.00	2.029	2.663	14.32	.00
Coverage	(.956)	(.767)	(.913)			(.863)	(.913)		
Content	2.241	2.056	2.913	12.44	.00	2.243	2.913	20.01	.00
Coherence	(.903)	(.780)	(.652)			(.867)	(.652)		
Argument	2.221	1.875	3.054	19.82	.00	2.043	3.054	31.37	.00
Quality	(1.053)	(.721)	(.845)			(.908)	(.845)		

Note. ** $p \leq .01$.

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Acknowledgments

The material is based upon work supported by the National Science Foundation under Grant No. 2247790 and Grant No. 2112532. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.